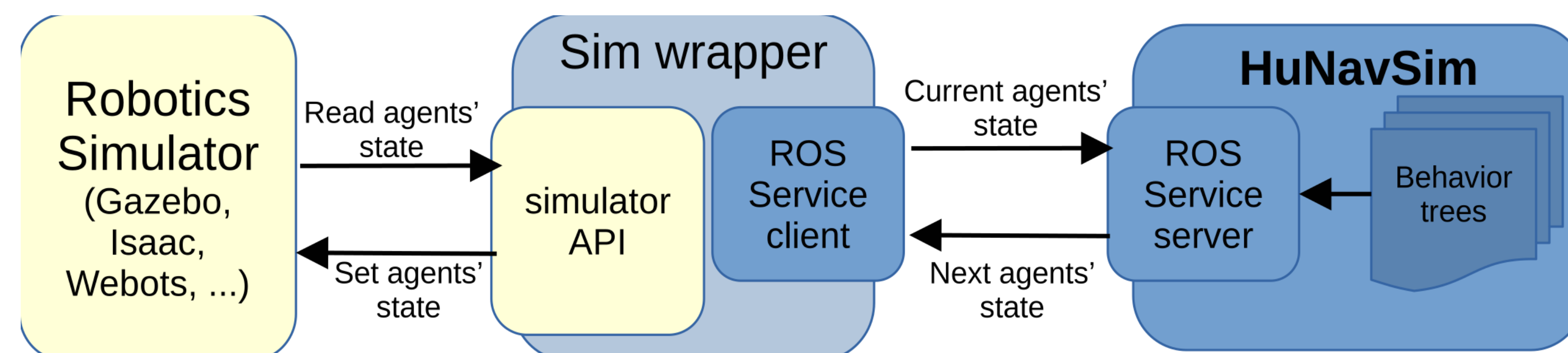


## HUMAN SIMULATION

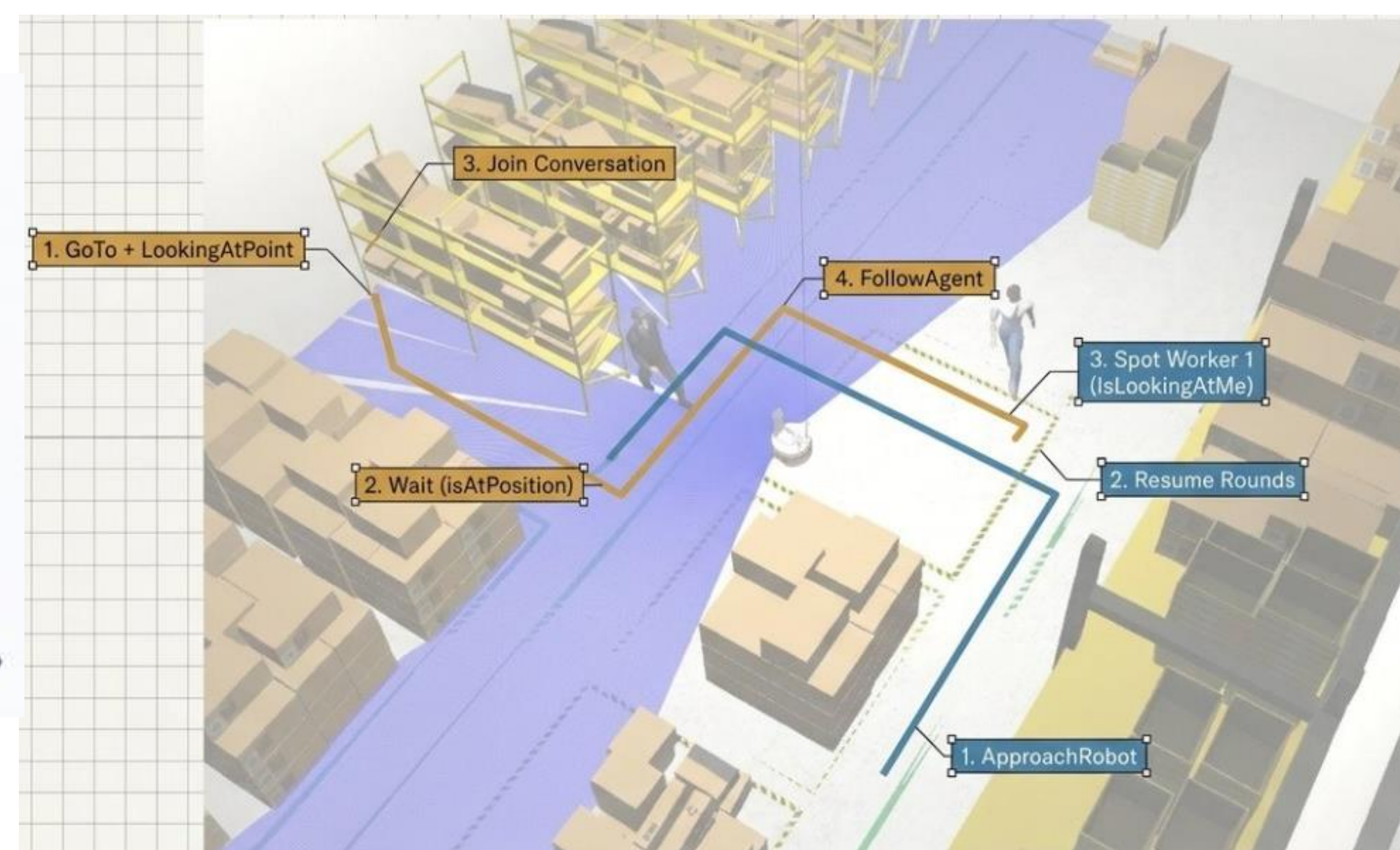
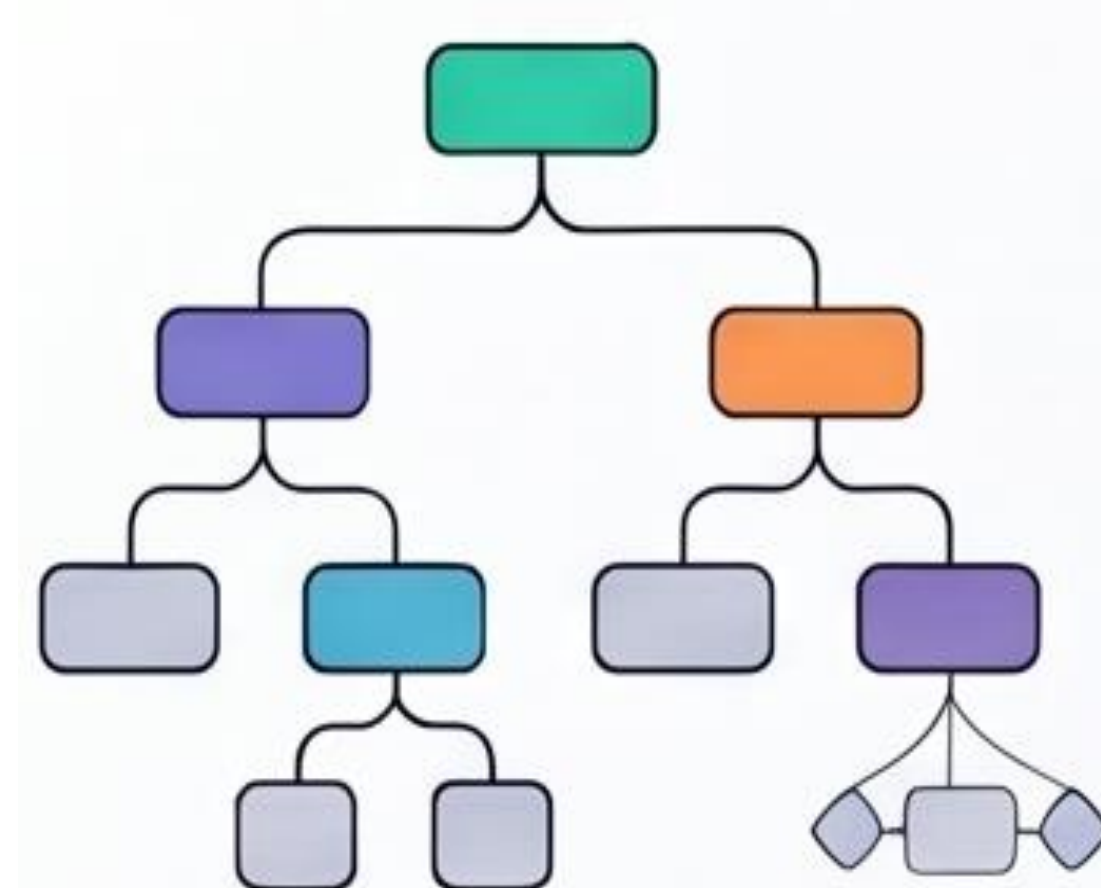
Open-source ROS 2 framework for human navigation simulation

### ROS2-based Architecture



### Behavior Tree Orchestration + "noisy" SFM

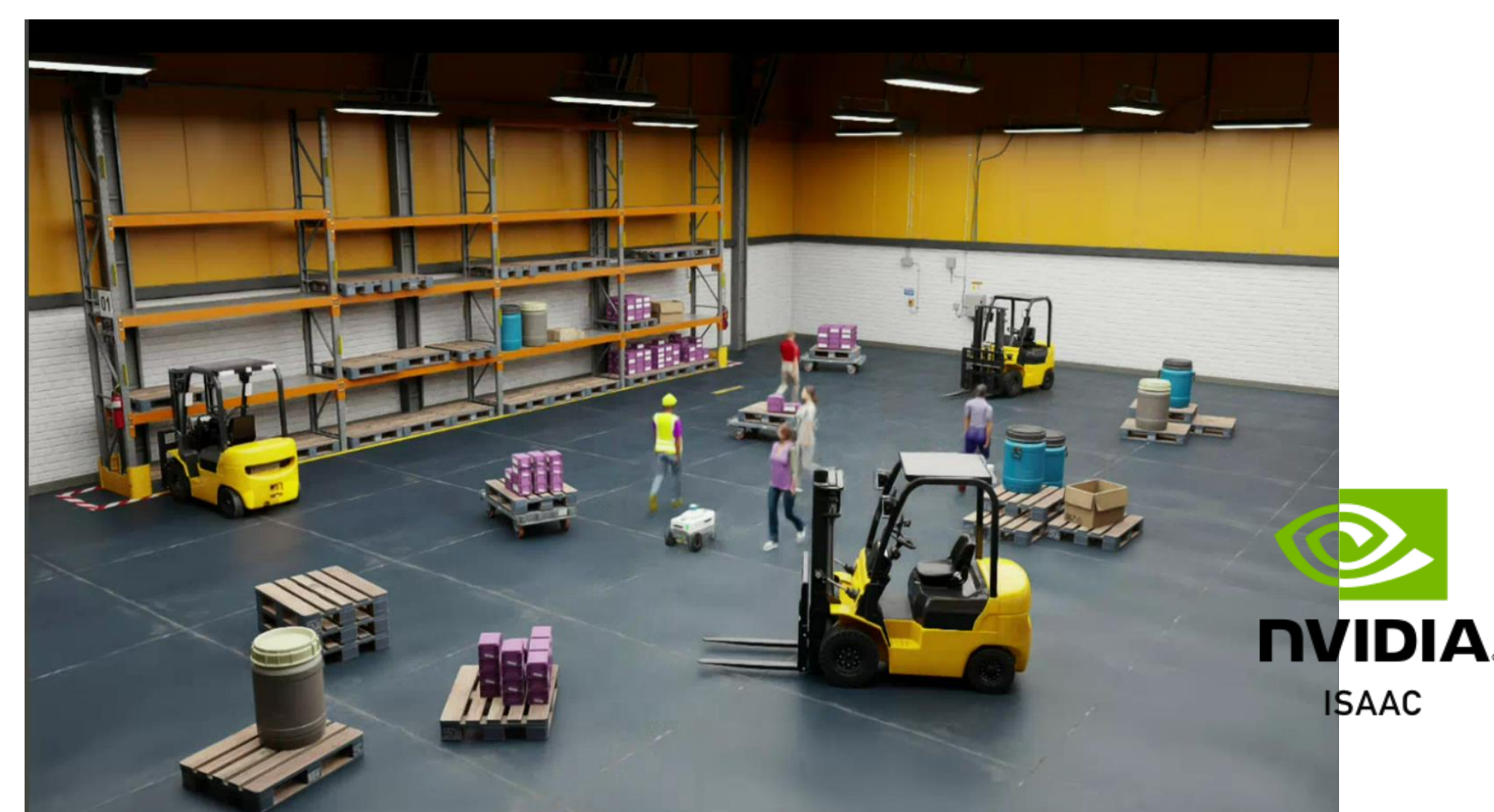
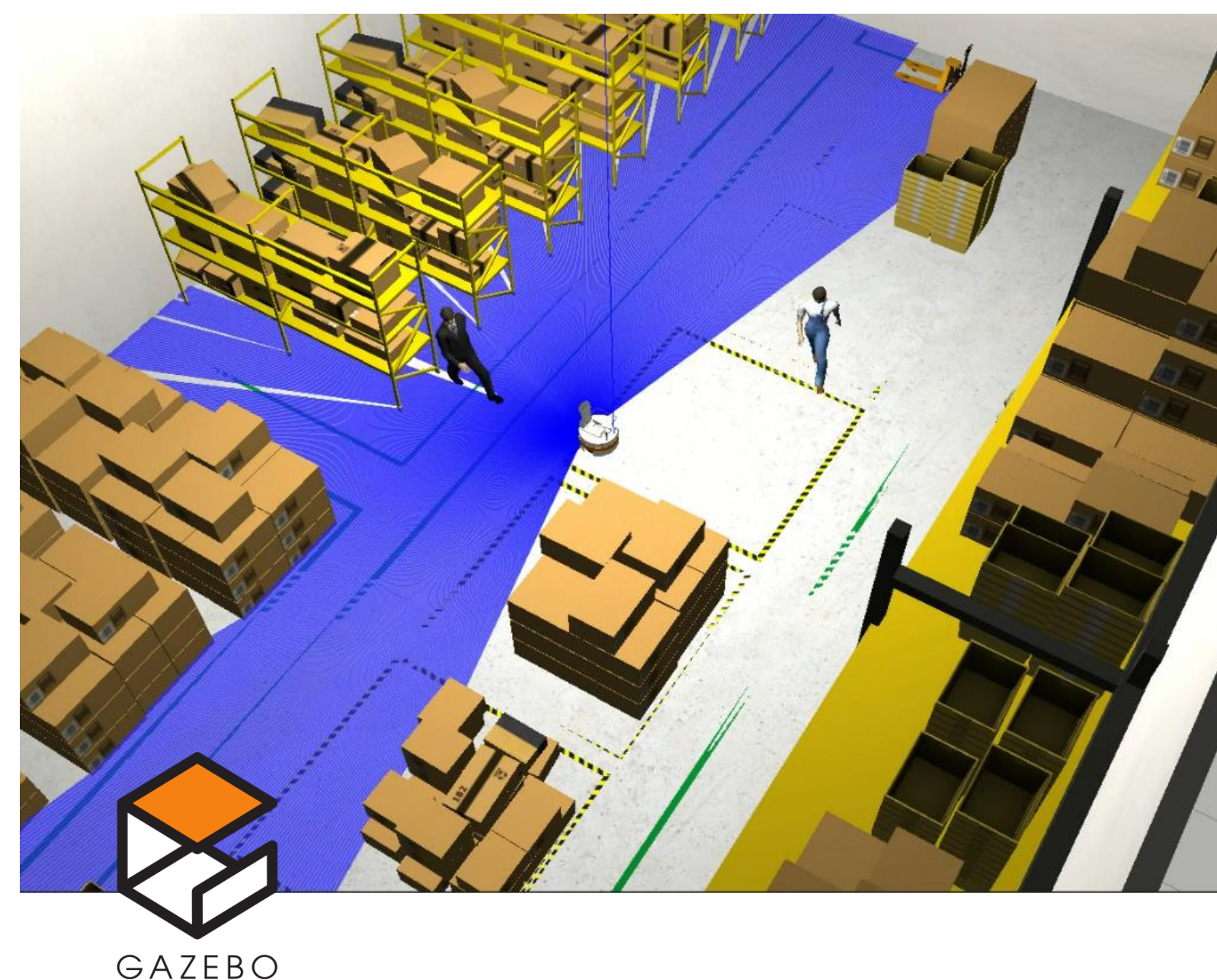
Complex social tasks beyond simple walking. Controlled noise in SFM to mimic natural human diversity



### Multi-simulator Integration

Wrappers for robotics engines like:

- NVidia Isaac Sim
- Gazebo
- Webots



## HUMAN-ALIGNED BENCHMARKING

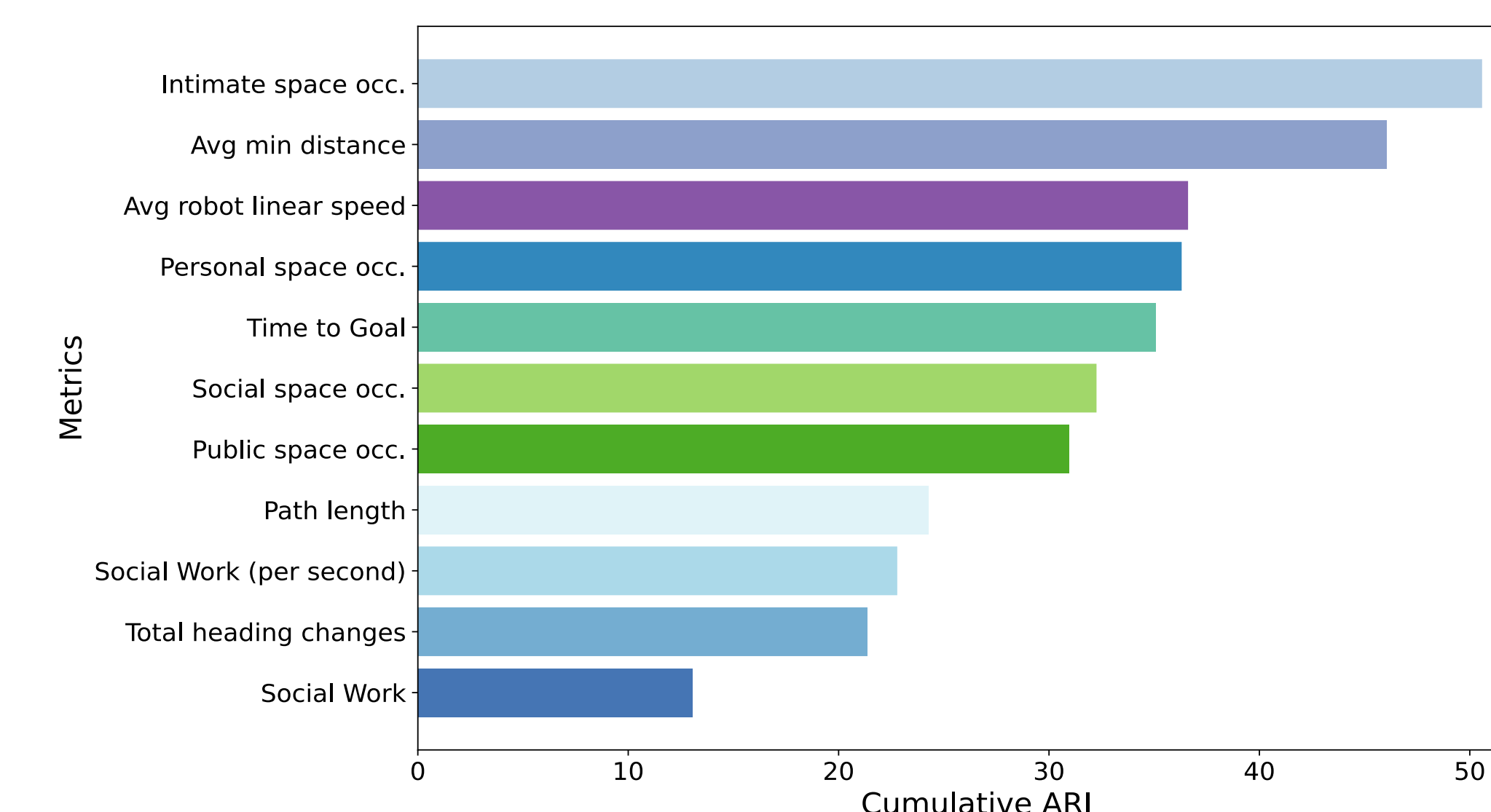
### Metrics suite

Compilation of 32 metrics for Human-aware navigation from the literature

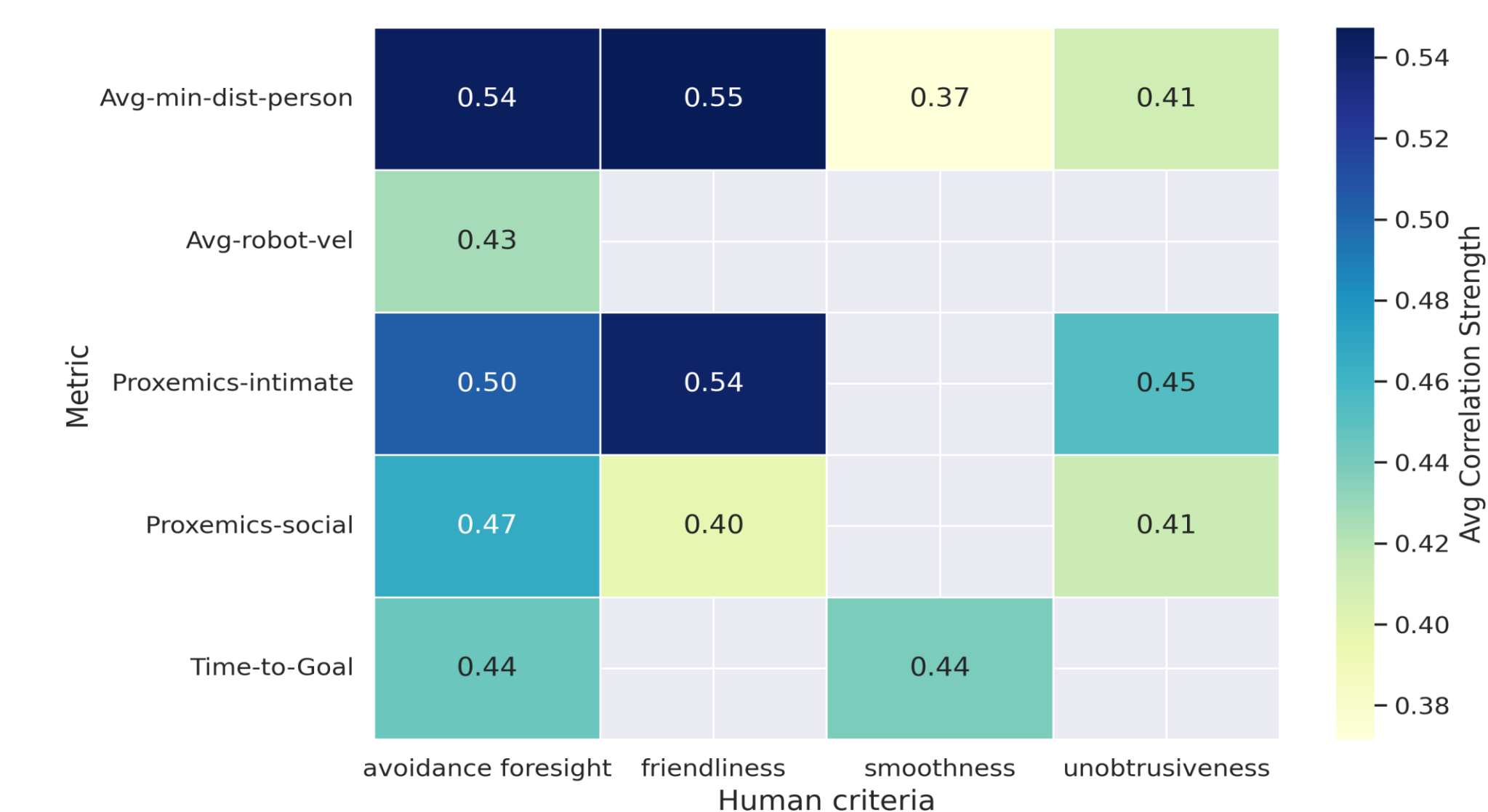
### Bridging the subjective gap

Goal: Identify quantitative metrics (QM) that statistically correlate with human-perceived assessments (HM) collected by survey.

### Unsupervised approach



### Correlation study



**Metrics Configuration**

Search metrics:

Total metrics: 28, Selected: 28

**Navigation Performance**

- ☒ Time Not Moving
- ☒ Final Distance To Target
- ☒ Time To Reach Goal
- ☒ Path Length
- ☒ Completed
- ☒ Minimum Distance To Target

**Social Behavior**

- ☒ Avg Closest Pedestrian Velocity
- ☒ Personal Space Intrusions
- ☒ Group Intimate Space Intrusions
- ☒ Avg Distance To Closest Person
- ☒ Minimum Distance To People
- ☒ Intimate Space Intrusions
- ☒ Social Space Intrusions
- ☒ Group Personal Space Intrusions
- ☒ Group Social Space Intrusions
- ☒ Avg Pedestrian Velocity

**Motion Analysis**

- ☒ Avg Acceleration
- ☒ Avg Robot Linear Speed
- ☒ Cumulative Heading Changes
- ☒ Avg Overacceleration
- ☒ Avg Robot Angular Speed

